# Jacob **Tutt**

PhD Candidate · Kavli Institute for Cosmology · Cavendish Laboratory · University of Cambridge

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## Summary


PhD researcher in machine learning and computational statistics at the Cavendish Laboratory and Kavli Institute for Cosmology, University of Cambridge. My research looks at how the latest statistical and probabilistic machine learning methods, combined with distributed, hardware-accelerated computing, can tackle previously intractable inference problems in physics. I apply this primarily to the statistical detection and characterisation of the cosmological 21-cm signal, but am broadly motivated by the challenge of scaling statistical and numerical methods, both algorithmically and computationally, to ever larger datasets, parameter spaces and physical systems.


## Education

### Ph.D. in Astrophysics — University of Cambridge, UK

*Sept 2025 - Present*

Developing scalable probabilistic machine learning and Bayesian inference methods to extract faint signals from noisy, high-dimensional and highly degenerate data in 21-cm cosmology. Selected for the first cohort of the Cavendish Frontiers Programme in AI for Scientific Discovery.

**Supervisors:** Prof Eloy de Lera Acedo and Dr Dominic Anstey

**Awards & Funding**

— **Harding Distinguished Postgraduate Scholarship** — Fully funded PhD scholarship supporting academically outstanding students across all disciplines.

— **Cavendish Scholarship** — Internal scholarship awarded by the Cavendish Laboratory to the top-ranked applicants to the PhD in Physics programme.

— **Compute Funding** — Primary researcher on three UKRI AIRR Gateway grants totalling up to 30,000 GPU hours, valued at c.£100,000.

— **Travel Funding** — UKRI Digital Research Infrastructure Travel Grant and Jesus College Postgraduate Research Fund (2026).

### MPhil in Data Intensive Science — University of Cambridge, UK

*Oct 2024 - Aug 2025*

Graduate research programme in statistical inference, machine learning and high-performance computing across DAMTP, the Department of Physics and the Institute of Astronomy. Thesis focused on unsupervised probabilistic modelling of the Milky Way's assembly history, alongside projects in cosmological inference from gravitational-wave detections and parameter-efficient adaptation of transformers.

**Supervisors:** Dr GyuChul Myeong, Prof Vasily Belokurov and Prof James Fergusson

**Grades**

— **High Distinction (85.2%)**

**Awards & Funding**

— **Jesus College Scholarship** — Awarded for outstanding academic achievement in the MPhil.

### BSc in Physics — Durham University, UK

*Oct 2021 - Sept 2024*

Took modules across the Departments of Mathematical Sciences and Physics, with research projects in the Institute for Computational Cosmology and the Centre for Extragalactic Astronomy.

**Grades**

— **First Class Honours, 4th in Degree Programme (80%)**

**Awards & Funding**

— **Durham Physics Award** — Recognised exceptional academic performance, including placing 4th in the degree programme.

— **Defence STEM Undergraduate Scholarship** — Awarded by the British Army for leadership performance at the Army Officer Selection Board.

## Publications

### First Author

**A Cross-Domain Simulator and Benchmark for Uncertainty Quantification in Global 21-cm Inference**

**J. L. Tutt**, H. T. J. Bevins, J. Alvey, et al. • *Sim2Science: ML with Imperfect Scientific Models, NeurIPS 2026*

**JAXMg: A Multi-GPU Linear Solver in JAX**

**J. Tutt**, R. Wiersema • arXiv:2601.14466 • *In review, Journal of Open Source Software*

**Towards End-to-End Bayesian Forward Models in Global 21-cm Cosmology: Surrogate Modelling and Marginalisation of Beam Uncertainty**

**J. L. Tutt**, D. J. Anstey, J. Cumner, et al. • arXiv:2608.18962 • *In review, MNRAS*

**Optimising Foreground Modelling for Global 21-cm Cosmology with GPU-Accelerated Nested Sampling**

**J. L. Tutt**, P. H. Sims, J. H. N. Pattison, et al. • arXiv:2603.13196 • *MNRAS (2026)*

**An Optimal Latent Space for Amortised Bayesian Uncertainty Quantification and Evidence Evaluation in Interferometric Visibilities**

**J. L. Tutt**, P. H. Sims, J. Burba, E. de Lera Acedo • *In preparation*

### Contributing Author

**Conditional Neural Bayes Ratio Estimation for Experimental Design Optimisation**

S. A. K. Leeney, T. Gessey-Jones, W. Handley, et al. (incl. **J. L. Tutt**) • arXiv:2603.26489 • *In review, IEEE TNNLS*

### Collaboration Papers

**Scalable Bayesian Data Curation for Next-Generation Radio Experiments**

S. A. K. Leeney et al., REACH Collaboration (incl. **J. L. Tutt**) • arXiv:2606.16525 • *In review, MNRAS*

## Professional Experience

**Army Reservist** — 4th Battalion, The Parachute Regiment, British Army *2022 – Present*

Undertaking the Res. Commissioning Course at Royal Military Academy Sandhurst to lead a very high readiness airborne platoon (30 soldiers).

**Investment Fellow** — Entrepreneur First *Nov 2026 – Present*

Identifying and assessing technical founders and deep-tech startups across Cambridge.

**Spring into Quant Finance** — G-Research *Mar – Apr 2026*

Selected from a graduate applicant pool with a $<$1% acceptance rate for G-Research's quantitative research spring internship programme.

## Research Software

**JAXMg** *Co-author and maintainer*

*Distributed multi-GPU and multi-node dense linear algebra for JAX (★ 114 GitHub stars)* [GitHub]

**REACH GPU Pipeline** *Main author and maintainer*

*GPU-accelerated forward modelling and Bayesian inference for global 21-cm cosmology in JAX* [Private] [GitHub]

**BayesEoR-v2** *Main author and maintainer*

*Amortised latent-space and hardware-accelerated Bayesian power-spectrum inference for 21-cm interferometry in JAX* [Private]

## Teaching

*M1 Machine Learning*, MPhil in Data Intensive Science, Graduate Supervisor — *Oct. 2026 – Present*

*C1 Research Computing*, MPhil in Data Intensive Science, Graduate Supervisor — *Oct. 2026 – Present*

*S1 Statistical Methods*, MPhil in Data Intensive Science, Graduate Supervisor — *Oct. 2025 – Present*

*S2 Advanced Statistical Methods*, MPhil in Data Intensive Science, Graduate Supervisor — *Oct. 2025 – Present*

*Part IA Physics*, Natural Sciences Tripos, Undergraduate Supervisor — *Oct. 2025 – Present*

## Talks

*Machine Learning Emulators for Fast Bayesian Inference in 21-cm Cosmology* — *Nov. 2026*
Cosmic Dawn Lab, University of Cambridge

*An Optimal Latent Space for Uncertainty Quantification and Evidence Evaluation in 21-cm Cosmology* — *Oct. 2026*
21-cm Meeting Series, University of Cambridge

*GPU-Accelerated Bayesian Forward Models: Uncertainty Quantification for Global 21-cm Cosmology* — *Sept. 2026*
IX Global 21-cm Workshop, University of Malta

*Modern Accelerators for Next-Gen Inference: End-to-End Uncertainty Quantification for Global 21-cm Cosmology* — *Sept. 2026*
REACH General Meeting, University of Malta

*Uncertainty Quantification in 21-cm Cosmology: Changing What Is Tractable with Modern Accelerators* — *Sept. 2026*
Kavli Astrophysics Symposium, University of Cambridge

*LLM-Era Compute for 21-cm Cosmology: Accelerating Bayesian Inference for the SKA Era* — *Mar. 2026*
AI/ML in Radio Astronomy Workshop, Royal Observatory Edinburgh

*Accelerating 21-cm Cosmological Inference for REACH with JAX/GPUs* — *Nov. 2025*
REACH Annual Meeting, University of Cambridge

## Skills

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|---|---|
| **Statistics** | Bayesian inference and model comparison, accelerated sampling, simulation-based inference, Gaussian processes |
| **Machine Learning** | Neural emulators, normalising flows, diffusion and flow-matching models, latent-variable models, NLP and fine-tuning |
| **Computing** | Multi-GPU and multi-node distribution, NVIDIA CUDA libraries, XLA FFI and custom JAX primitives |
| **Languages** | Python, C++, CUDA, JAX, PyTorch, TensorFlow, Bash, Git, LaTeX |